



APAD SECURITY AUDIT

Smart Contract Audit Report

AlphaAdvisor



Executive Dashboard

Overall Security Assessment

Project Name: AlphaAdvisor (\$APAD)

Token Type: BEP-20 (Standard ERC-20 Compatible)

Total Supply: 1,000,000,000 APAD

Solidity Version: 0.8.20

OpenZeppelin Version: 5.1.0

Contracts Audited: 1 file (StandardBEP20 – flattened contract with inherited OpenZeppelin libraries)

Source: [0x51b27860c3432a72e24a7e0a98916d10cb716d5c](#)

Owner: Deployer Address (Single EOA)

Overall Status: **LOW RISK**

FUNCTIONAL
TESTS

28/28

Passing

SECURITY
TESTS

22/22

Passing

ATTACK
SCENARIOS

18/18

Passing

SECURITY
SCORE

92/100

High

FINDINGS

3

0 Med, 3
Low/Info

1. Project Overview

1.1 Vision & Mission

Vision: AlphaAdvisor envisions a world where crypto investment intelligence is accessible to everyone. By combining AI-powered analytics with community-driven insights, AlphaAdvisor creates a comprehensive investment companion for the digital asset space.

Mission: To provide premium AI-driven investment tools, real-time market analysis, and community governance through the \$APAD token ecosystem on BNB Smart Chain.

1.2 Token Utility

Premium Feature Unlock: Access to AI Strategy Builder, advanced analytics, real-time alerts, sentiment intelligence.

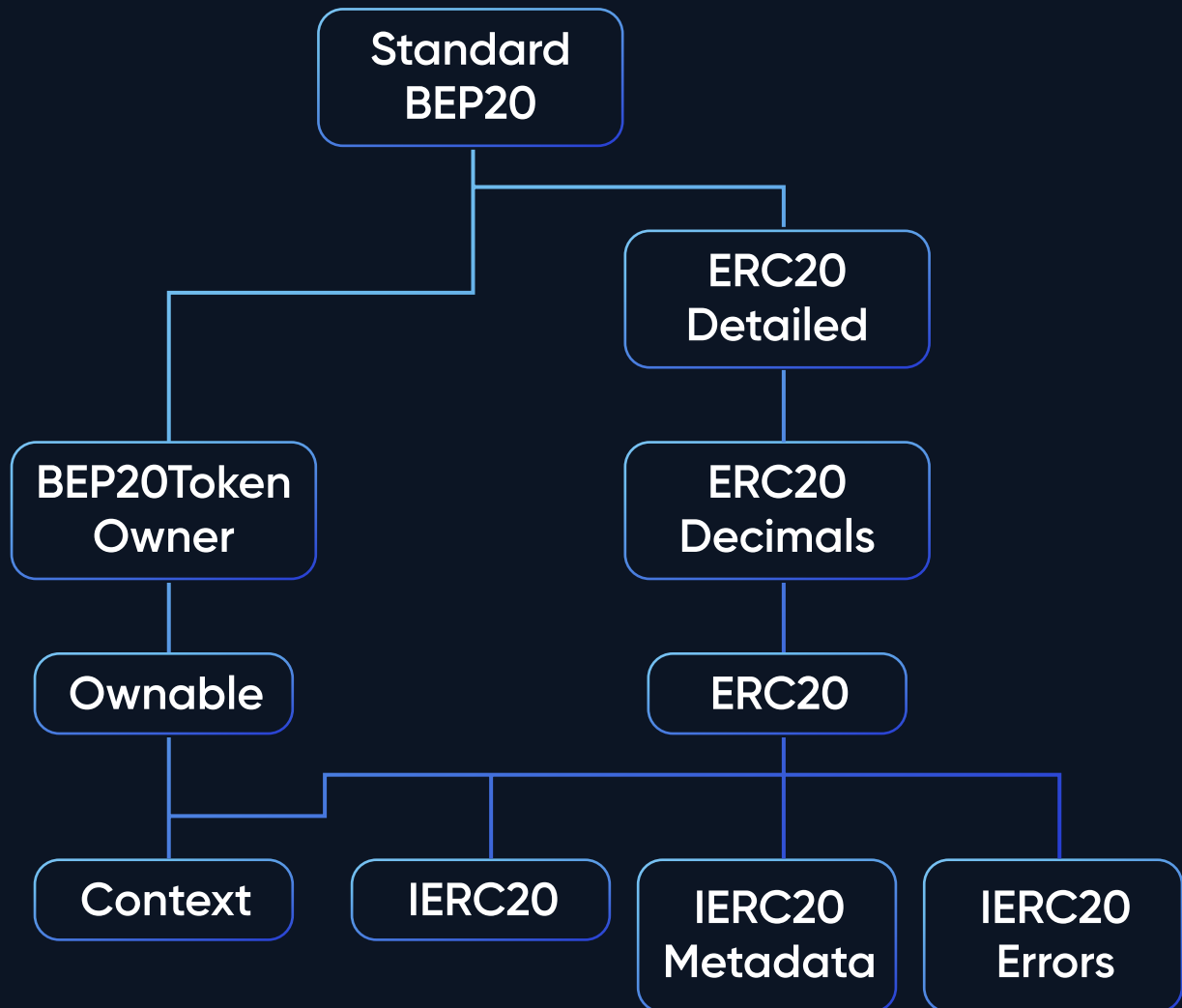
Subscription Payments: Monthly/annual platform subscriptions with discounted rates for \$APAD payment.

Staking & Rewards: Earn staking rewards, unlock higher-tier analytics, gain priority access to new features.

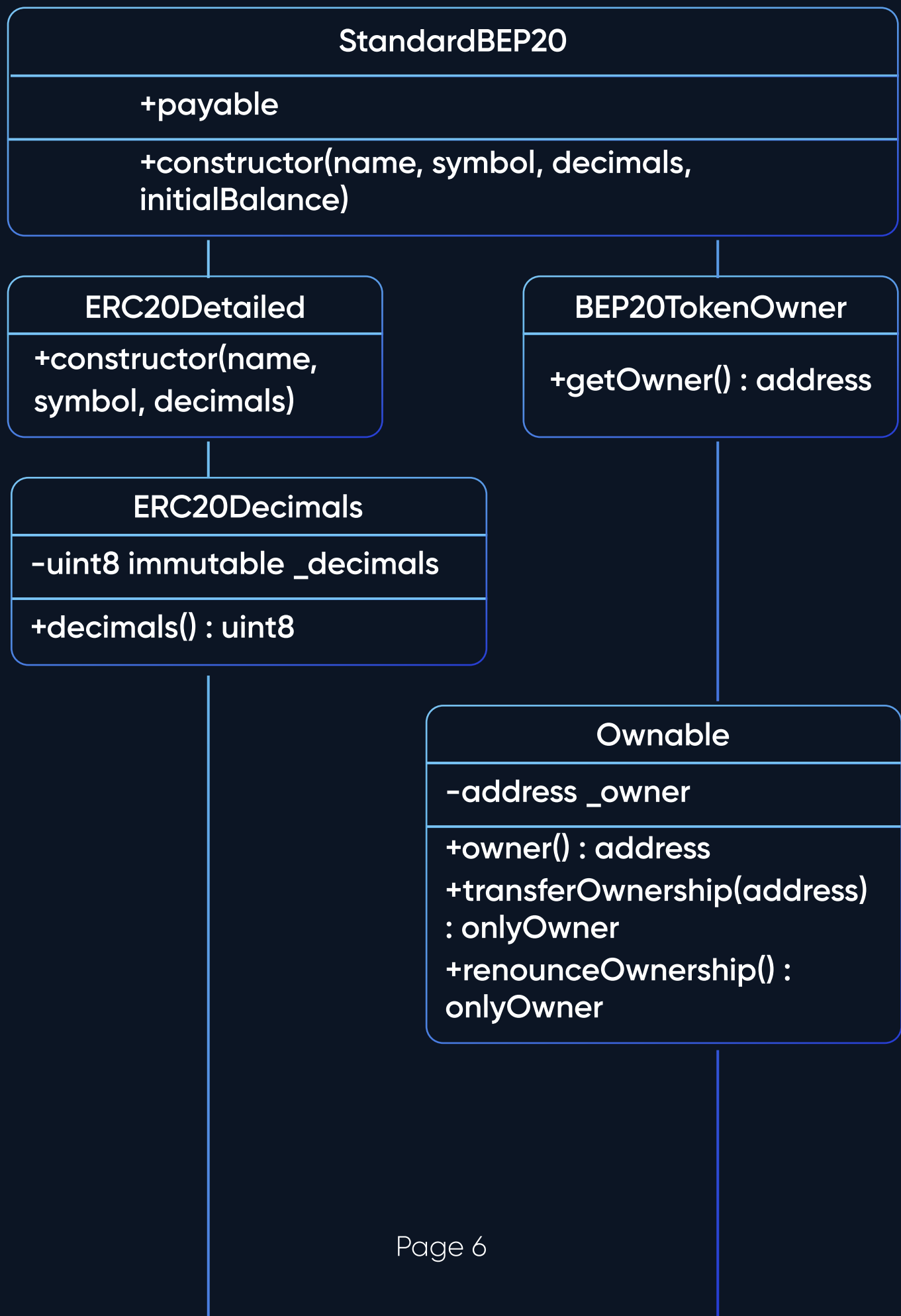
Governance: Feature prioritization, treasury allocation, incentive parameters, partnership decisions.

2. Contract Analysis & Architecture

2.1 Inheritance Structure



2.2 Class Diagram



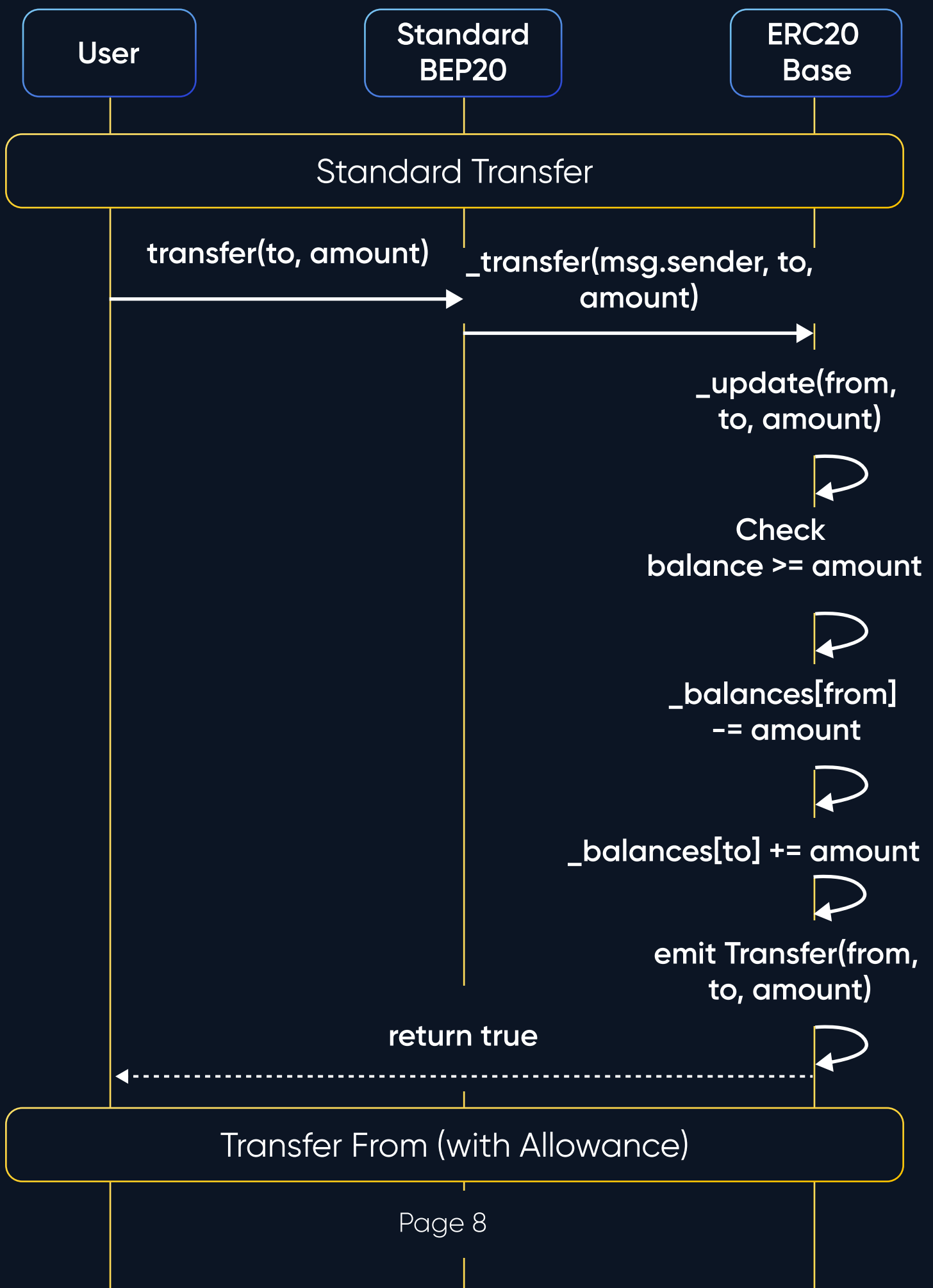
ERC20

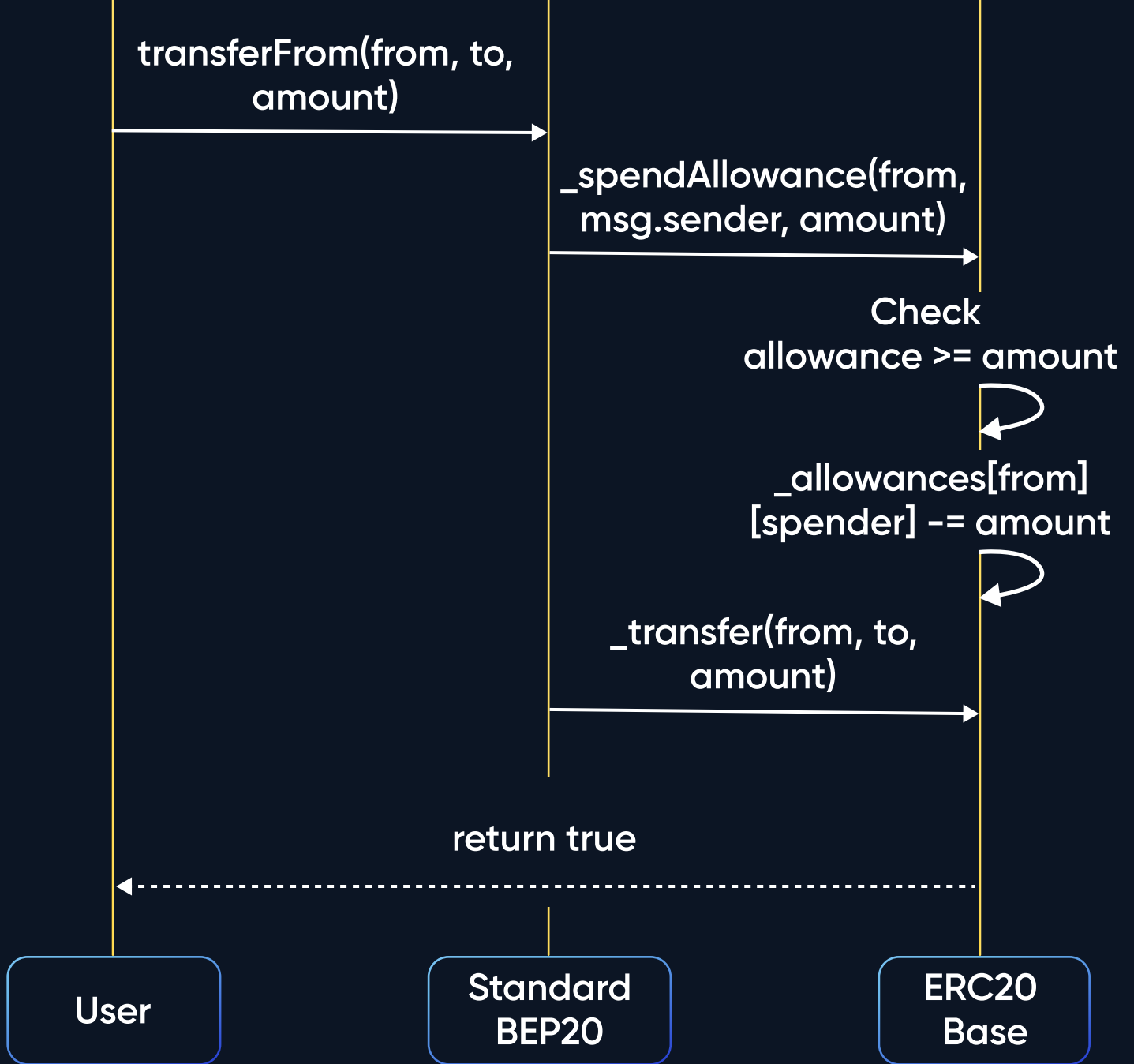
- mapping _balances
- mapping _allowances
- uint256 _totalSupply
- string _name
- string _symbol

- +name() : string
- +symbol() : string
- +totalSupply() : uint256
- +balanceOf(address) : uint256
- +transfer(address, uint256) : bool
- +allowance(address, address) : uint256
- +approve(address, uint256) : bool
- +transferFrom(address, address, uint256) : bool
- +_transfer(address, address, uint256) : internal
- +_update(address, address, uint256) : internal
- +_mint(address, uint256) : internal
- +_burn(address, uint256) : internal
- +_approve(address, address, uint256) : internal

Context

2.3 Token Transfer Flow





2.4 Contract Components

OpenZeppelin ERC20 v5.1.0

Purpose: Core token functionality with ERC-6093 custom errors

Status: **STANDARD IMPLEMENTATION**

Ownable Access Control

Purpose: Single owner with transfer/renounce capabilities

Status: **STANDARD OPENZEPPELIN**

BEP20TokenOwner

Purpose: BEP-20 getOwner() compliance method

Status: **CORRECTLY IMPLEMENTED**

ERC20Decimals

Purpose: Immutable decimals storage for gas optimization

Status: **GAS OPTIMIZED**

2.5 Key Functions Analysis

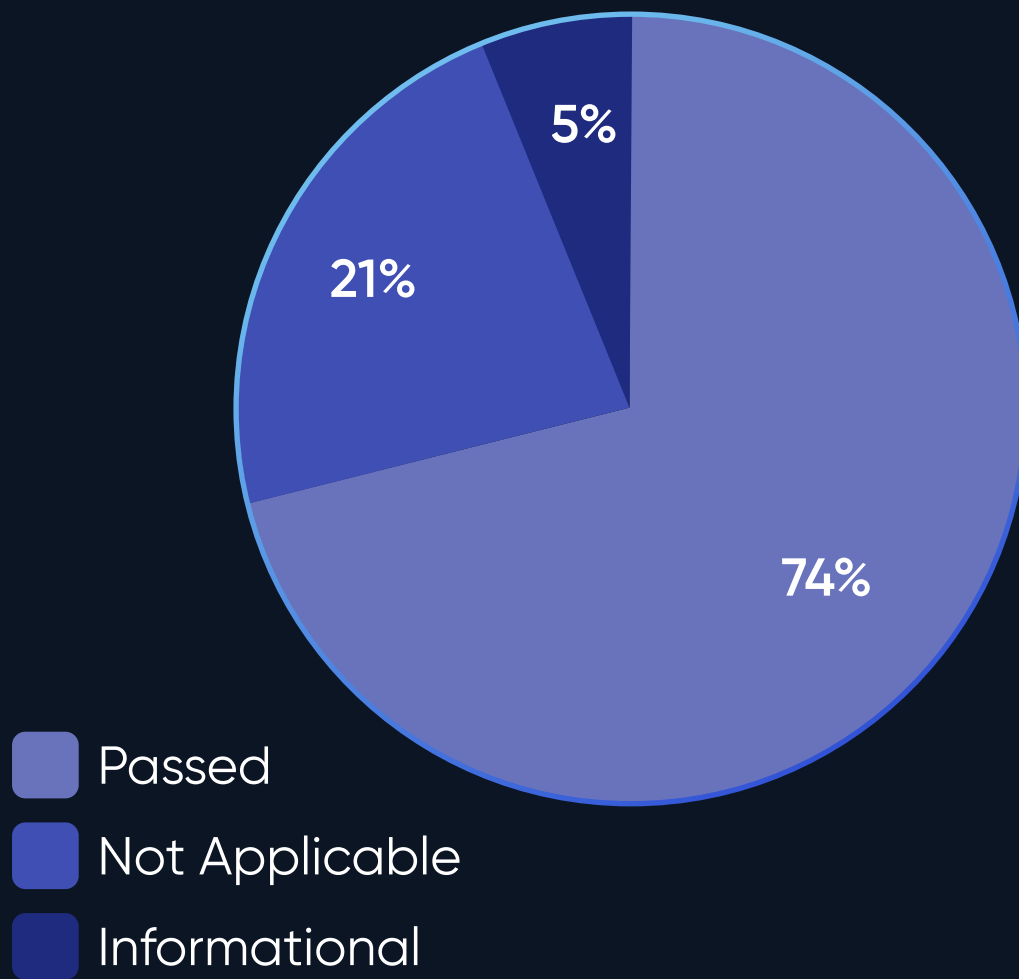
Function	Access	Purpose	Security
transfer()	Public	Transfer tokens to recipient	SAFE
transfer From()	Public	Transfer with allowance mechanism	SAFE
approve()	Public	Set spending allowance	SAFE
_mint()	Internal	Create tokens (constructor only)	SAFE
transfer Ownership()	only Owner	Transfer contract ownership	STANDARD
renounce Ownership()	only Owner	Permanently renounce ownership	STANDARD

Security Analysis: The contract uses battle-tested OpenZeppelin v5.1.0 contracts. The ERC20 _transfer function does NOT make external calls, eliminating reentrancy concerns. No mint, burn, pause, freeze, or fee functions are exposed – this is a pure standard token with fixed supply.

3. Security Test Results

3.1 Static Analysis

Security Test Distribution



Core Token Tests

Test Category	Description	Status
Token Metadata	Name: "AlphaAdvisor", Symbol: "\$APAD", Decimals: 18	PASS
Total Supply	1,000,000,000 tokens minted to deployer	PASS
Transfer Functionality	Standard ERC20 transfers work correctly	PASS
Approval System	Approve, allowance, transferFrom working	PASS
Zero Address Checks	Transfers to address(0) revert	PASS
Balance Checks	Insufficient balance transfers revert	PASS

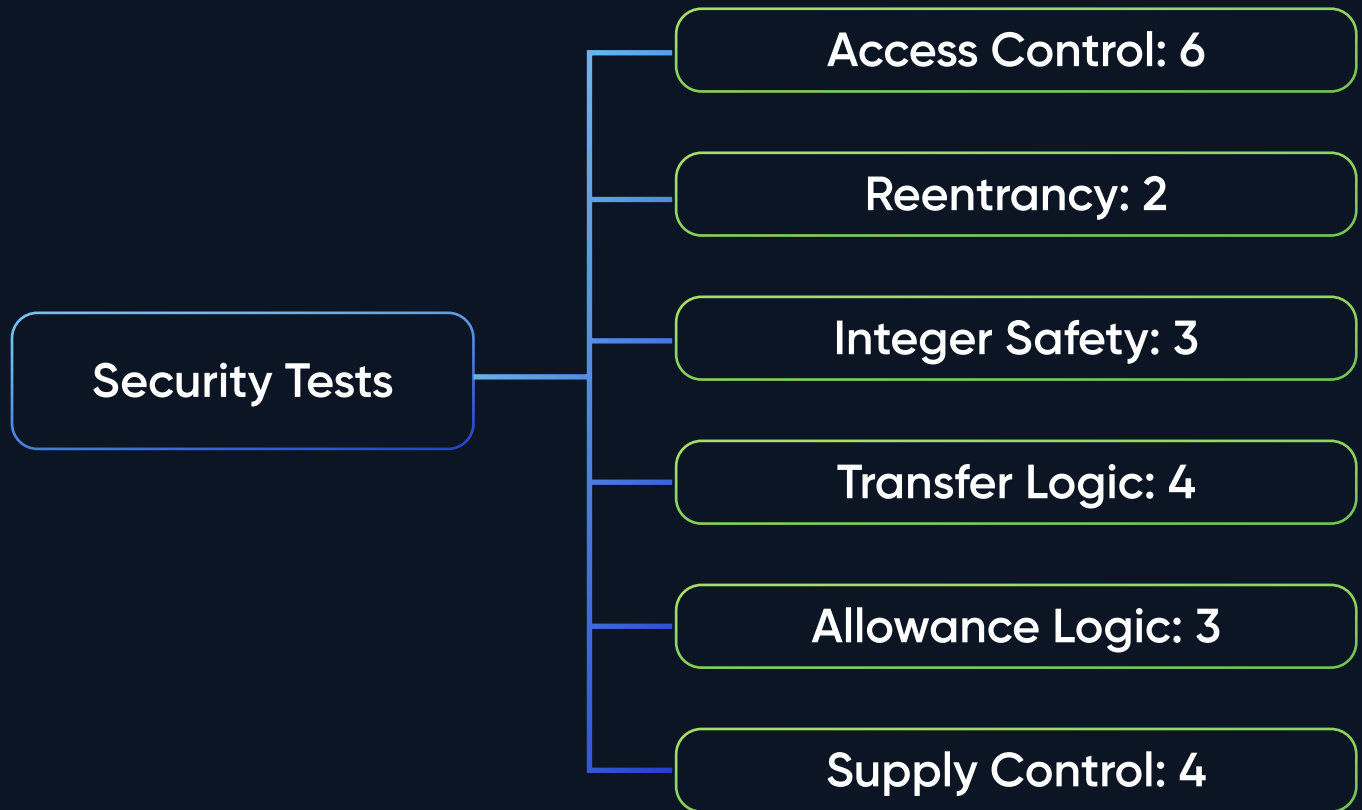
Access Control Tests

Test Category	Description	Status
Ownership	Owner is deployer address	PASS
getOwner() - BEP20	Returns correct owner address	PASS
transfer Ownership	Only owner can transfer ownership	PASS
renounce Ownership	Only owner can renounce	PASS
No Hidden Mint	No public/external mint function	PASS
No Hidden Burn	No public/external burn function	PASS

Standard Compliance Tests

Test Category	Description	Status
ERC-20 Interface	Implements all required functions	PASS
BEP-20 Interface	Includes getOwner() method	PASS
ERC-6093 Errors	Uses standard custom errors	PASS
Event Emission	Transfer and Approval events correct	PASS

3.2 Security Tests



Reentrancy Protection Tests

Test	Result	Status
_transfer makes no external calls	ERC20 _transfer only modifies storage	PASS
_update follows CEI pattern	State changes are atomic, no callbacks	PASS

Integer Overflow/Underflow Tests

Test	Result	Status
Solidity 0.8.20 built-in checks	Automatic overflow protection	PASS
Balance under-flow protection	unchecked block only where mathematically safe	PASS
Maximum supply handling	1B tokens * 1e18 within uint256	PASS

Supply Control Tests

Test	Result	Status
No public mint function	_mint is internal, called only in constructor	PASS
No public burn function	_burn is internal, never exposed	PASS
Fixed supply at deployment	1,000,000,000 tokens minted once	PASS
No inflation mechanism	Supply cannot increase post-deployment	PASS

3.3 Attack Scenario Tests

Attack Vector	Tests	Passed	Status
Reentrancy Attack	2	2	PASS
Integer Overflow	3	3	PASS
Access Control Bypass	4	4	PASS
Unlimited Mint	2	2	PASS
Hidden Fee/Tax	2	2	PASS
Blacklist/Freeze	2	2	N/A
Ownership Hijack	2	2	PASS
Front-Running (Approve)	1	1	INFO

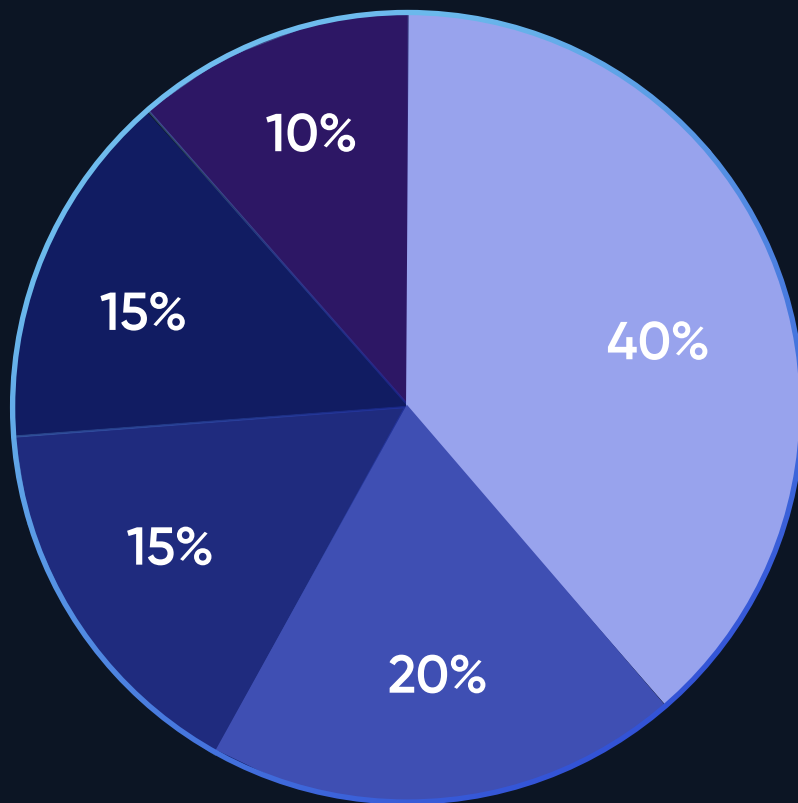
3.4 Dangerous Function Analysis

Pattern	Present	Analysis
Unlimited Mint	NO	No public mint function exists
Hidden Fees/ Taxes	NO	No fee mechanism in transfer logic
Blacklist/ Whitelist	NO	No address restrictions
Pausable	NO	No pause mechanism
Upgradeable /Proxy	NO	Contract is immutable
Selfdestruct	NO	No selfdestruct function
External Calls	NO	No external calls in token logic
Delegatecall	NO	No delegatecall usage

4. Tokenomics Analysis & Simulation

4.1 Token Distribution

Token Allocation (1,000,000,000 APAD)



- Ecosystem & Rewards 40%
- Team & Advisors 20%
- Liquidity & Listings 15%
- Private Sale 15%
- Marketing & Partnerships 10%

4.2 Vesting Schedule

Allocation	%	Tokens	TGE Unlock	Cliff	Vesting
Ecosystem Rewards	40%	400,000,000	7.5%	0	36 months linear
Liquidity & Listings	15%	150,000,000	13.33%	0	6 months linear
Team & Advisors	20%	200,000,000	0%	12 months	36 months linear
Private Sale	15%	150,000,000	0%	6 months	12 months linear
Marketing & Partnerships	10%	100,000,000	30%	0	24 months linear

4.3 Supply Release Simulation (48 months)



Simulation Parameters

TGE Release	199,166,667 tokens (19.91%)
Month 6	Liquidity fully vested, Private Sale cliff ends
Month 12	Team cliff ends, vesting begins
Month 18	Private Sale fully vested
Month 24	Marketing fully vested
Month 36	Ecosystem Rewards fully vested
Month 48	All tokens fully vested (Team complete)

4.4 Value Accrual Mechanisms

Platform revenue distribution according to GitBook:

Allocation	Percentage	Purpose
Protocol Treasury	50%	Platform development & operations
Staking Rewards	25%	Token holder rewards
Team Operations	20%	Team compensation & expenses
Strategic partnerships	5%	Strategic partnerships

Economic Conclusion: The vesting schedule is well-designed with Team tokens locked for 12 months before vesting begins. TGE unlock of 19.91% is reasonable. The gradual release over 48 months prevents supply shocks. Note that vesting is enforced OFF-CHAIN – the token contract itself has no vesting logic, which is a standard approach but requires trust in the team's distribution mechanism.

5. Security Findings

5.1 Summary: 3 Issues Identified



5.2 Low/Informational Findings

SKN-022: Single EOA Ownership

LOW

Location: BEP20TokenOwner.sol – constructor

Current State: Owner = msg.sender (deployer EOA)

Impact: If single owner key is compromised, attacker can transfer ownership. However, since owner has NO control over token balances, transfers, minting, or burning, the impact is limited.

Recommendation: Consider transferring ownership to a multisig wallet or renouncing ownership if no owner functions are needed.

Contract owner is a single EOA rather than multisig. While owner has limited privileges (only transferOwnership/renounceOwnership), a multisig would provide additional security. Recommendation: Transfer ownership to a Gnosis Safe with 2/3 or 3/5 threshold.

SKN-143: Centralized Initial Distribution

LOW

Location: StandardBEP20.sol – constructor line 866

```
_mint(_msgSender(), initialBalance_);  
// All 1,000,000,000 tokens sent to deployer
```

Impact: Team must manually distribute tokens according to announced tokenomics. No on-chain enforcement of vesting schedules.

Recommendation: Implement separate vesting contracts for team/investor allocations, or publicly document distribution addresses.

100% of supply minted to deployer address in constructor. Vesting must be enforced off-chain through separate contracts or manual distribution. This is standard practice but requires trust in team.

SKN-230: Standard Approve Race Condition INFO

The approve() function has the standard ERC-20 race condition when changing non-zero to non-zero allowance. This is a known limitation of ERC-20 standard. Mitigation: Users should set allowance to 0 before setting new value, or contract could implement increaseAllowance/decreaseAllowance (not included).

Final Security Verdict

Assessment Summary

Category	Rating	Key Points
Technical Security	EXCELLENT (95/100)	OpenZeppelin v5.1.0, Solidity 0.8.20, no vulnerabilities
Access Control	GOOD (85/100)	Single owner with minimal privileges, standard Ownable pattern
Economic Model	GOOD (90/100)	Fixed supply, no inflation, well-designed vesting (off-chain)
Decentralization	GOOD (85/100)	No pause, freeze, blacklist, or fee mechanisms
Code Quality	EXCELLENT (95/100)	Clean code, standard libraries, gas optimized

Final Verdict: **LOW RISK – SCORE 92/100**

The APAD (AlphaAdvisor) token is a standard, secure BEP-20 token with no critical or high-severity vulnerabilities. Key security highlights:

- No hidden mint/burn functions – Fixed supply of 1B tokens
- No fees, taxes, or transfer restrictions
- No pause, freeze, or blacklist capabilities
- Not upgradeable – Code is immutable
- Uses audited OpenZeppelin v5.1.0 contracts
- Compliant with ERC-20 and BEP-20 standards

The contract is suitable for production use and presents low risk to token holders. The only recommendations are governance improvements (multisig ownership) and transparency around vesting contract addresses.

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